

MISSOURI


MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Section 2.2

The Inverse of a Matrix

The Determinant of a 2×2 Matrix

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. The determinant of A is

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example 1

Calculate the determinant of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

The Inverse of a Matrix

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C such that

$$CA = I_n \text{ and } AC = I_n$$

C is the inverse of A and we usually write
 $C = A^{-1}$

An invertible matrix is called nonsingular.

A noninvertible matrix is called singular.

Example 2

Compute the products CA and AC given

$$A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix} \text{ and } C = \begin{bmatrix} -2 & 1 \\ \frac{7}{2} & -\frac{3}{2} \end{bmatrix}.$$

The Inverse of a 2×2 Matrix

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

If $\det A \neq 0$, then A is invertible.

Specifically,

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $\det A = 0$, then A is not invertible.

Example 3

Calculate the inverse of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solving Systems Using Inverses

If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Example 4

Solve the system.

$$\begin{aligned} x + 2y &= 5 \\ 3x + 4y &= 6 \end{aligned}$$

Facts About Invertibility

If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

If A and B are invertible $n \times n$ matrices, then so is AB ; specifically,

$$(AB)^{-1} = B^{-1}A^{-1}$$

Example 5

Prove that $(AB)^{-1} = B^{-1}A^{-1}$ by calculating $(AB)(B^{-1}A^{-1})$

and

$$(B^{-1}A^{-1})(AB)$$

Example 6

Suppose A , B , and C are invertible $n \times n$ matrices. Show that ABC is also invertible by producing a matrix D such that

$$(ABC)D = I$$

and

$$D(ABC) = I$$

Facts About Invertibility

The product of invertible $n \times n$ matrices is invertible, and their inverse is the product of their inverses in the reverse order.

If A is an invertible matrix, then so is A^T .
Specifically,

$$(A^T)^{-1} = (A^{-1})^T$$

Elementary Matrices

An elementary matrix is a matrix obtained by performing a single elementary row operation on an identity matrix.

Example 7

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$.

Perform the replacement operation

$$-4R_1 + R_2 \rightarrow R_2$$

on A to get B and on I_3 to get E .

Then, multiply EA .

Elementary Matrices

If an elementary row operation is performed on an $m \times n$ matrix A , the resulting matrix can be written as EA , where the $m \times m$ matrix E is created by performing the same row operation on I_m .

Elementary Matrices

Each elementary matrix E is invertible, and the inverse of E is itself an elementary matrix.

Theorem

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

Example 8

Prove this theorem.

Algorithm for Finding A^{-1}

To find A^{-1} , row reduce the augmented matrix
 $[A \quad I]$

If A is row equivalent to I (and thus A is invertible), then $[A \quad I]$ is row equivalent to
 $[I \quad A^{-1}]$

Example 9

Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$
